$$R_t^s|PR_{t-12,t-1}^s = \begin{cases} +r_t^s \frac{40\%}{\sigma_{t-1}^s}, & PR_{t-12,t-1}^s \geqslant 0 \\ -r_t^s \frac{40\%}{\sigma_{t-1}^s}, & PR_{t-12,t-1}^s < 0 \end{cases}, \tag{4}$$

$$R_t^s|P_{t-12,t-1}^s, q = \begin{cases} +r_t^s \frac{40\%}{\sigma_{t-1}^s}, & P_{t-12,t-1}^s > q \\ -r_t^s \frac{40\%}{\sigma_{t-1}^s}, & P_{t-12,t-1}^s < q \end{cases}. \tag{5}$$

Finally, for a universe of S assets, the portfolio return is calculated as:

$$R_t^p = \frac{1}{S} \sum_{s=1}^{S} R_t^s|P_{t-12,t-1}^s, q, \tag{6}$$

where R_t^s is the risk-adjusted return of each trend-following strategy for each individual instrument, using the above volatility scaling method, and R_t^p is the mean of all the R_t^s.

3. Time Series Reversal

According to the results of the OLS pooled regressions in Moskowitz et al. (2012)[9], time series reversal pattern is statistically insignificant. However, they have only used single month return and return sign as the explanatory variables, which are not exactly equivalent to the blocks that form time series momentum signals. Time series momentum considers the past month returns as a whole, for instance, period return over the past 12 months. By contrast, Papailias et al. (2021) examined the predictive power of RSM strategies by using a lagged average of return signs P. P covers all the return signs in the look-back period and can directly form the RSM signal. Their results suggest that a significant reversal pattern exists following the short-term momentum effect.

In this section, we extend the above studies by adding more explanatory variables to investigate the predictive power of TSM and RSM signals. First, as in Moskowitz et al. (2012), we regress the volatility-scaled return r_t^s/σ_{t-1}^s on the lagged signal

[9]See Figure (1) in their paper.